

HANAN HUSSAIN

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EDUCATION

University of Saskatchewan

Saskatoon, SK

Bachelor of Science in Computer Engineering

June 2026

Relevant Coursework: Software Engineering, Digital Design, Embedded Systems, Verification, Computer Networks

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Verilog, JavaScript (React, Node.js), HTML/CSS, SQL

Frameworks & Tools: Django, Next.js, Tailwind CSS, Linux, Bash, Git, MATLAB, Quartus, ModelSim

Hardware & Embedded: Raspberry Pi, Arduino, ARM Microcontrollers, FPGA (Intel), Logic Design, FPGA Programming

Soft Skills: Strong communication, teamwork, adaptability, attention to detail, time management, problem solving

PROJECTS

Two-Way Portable English-to-ASL Translator (Capstone)

Sept 2025 – Apr 2026

Raspberry Pi 5, Python, MediaPipe, VOSK STT, C, Linux, I2S/I2C/CSI, Edge Inference

- Designed and integrated a real-time bidirectional ASL–English translation system on Raspberry Pi 5; achieved end-to-end latency under 2 seconds with fully offline operation
- Developed a key-point-based gesture recognition pipeline using Python and MediaPipe; implemented landmark extraction, inference logic, and token mapping
- Architected a modular multi-threaded system managing camera, microphone, display, and speaker subsystems concurrently on a single embedded platform
- Implemented speech-to-text (VOSK) and text-to-speech pipelines with grammar conversion and ASL token sequencing, all running on-device
- Performed system-level verification including timing analysis, latency benchmarking, functional validation, and integration debugging

KawaKraft – Full-Stack E-Commerce Platform

2025

Django, Next.js, Tailwind CSS, PostgreSQL, REST API, GitHub

- Designed and launched a full-stack e-commerce website with secure REST APIs for product management, authentication, and customer orders
- Built responsive UI/UX with Next.js and Tailwind CSS; implemented version-controlled CI workflow using GitHub

Travel Planning Web Application

2024

Django, Python, SQL, relational database

- Developed a Django-based collaborative travel planning application with booking and reservation management backed by a relational database

8-bit Microprocessor (Verilog HDL)

2024

Verilog, ModelSim, Quartus, Intel FPGA

- Designed a modular 8-bit microprocessor in Verilog with sequencer, instruction decoder, and ALU; validated on Intel FPGA using Quartus synthesis and ModelSim simulation

CERTIFICATIONS

Google Cloud Fundamentals – Core Infrastructure (2025)

GitHub Foundations Certificate (2024)

AWS Educate – Cloud Computing (2023)

VOLUNTEER & ACTIVITIES

University of Saskatchewan Engineering Society – Events Team | Assisted organizing community outreach and networking events for engineering students